Phase 1: 1.0 Business Understanding

Gordon Oboh

Department of Data Science, Monroe College, King Graduate School

CS703-152W: Applied Data Science Project

Professor Nicholas Nardi

May 9, 2026

Phase 1: 1.0 Business Understanding

Task 1.1: Determine Business Objectives

The primary objective of this project is to support real estate investors and property managers operating in the Southern USA by identifying which apartment amenities most significantly influence rental price. Understanding amenity importance enables strategic property improvements and competitive pricing decisions in high-growth Southern markets.

Target Audience

- Real estate investors seeking to maximize return on amenity upgrades
- Property managers aiming to set competitive rental prices relative to local Southern market conditions

Business Questions

- Which amenities have the strongest impact on rental pricing?
- How does amenity importance vary across geographic regions?
- Which amenity upgrades yield the highest price premium relative to cost?
- Do certain amenity bundles (e.g., parking and laundry) have compounding effects on price?

Success Criteria

The project succeeds when a model can identify the top amenity drivers of rental price, producing rankings that are actionable for renovation prioritization and pricing strategy in Southern apartment markets.

Task 1.2: Assess the Situation

Stakeholder Analysis

Investors and property managers share a common anchor: understanding what drives property value. Investors seek to maximize return on capital allocated to amenity upgrades. Property managers need to price units competitively against comparable listings in the same Southern city. Both groups benefit from a clear, ranked understanding of which amenities move rental price.

Data Source and Inventory

The dataset was sourced from Kaggle (Craigslist scrape, January 2020): 384,977 rows, 22 columns, 558 MB. The target variable is `price` (monthly rental price in USD). Features include square footage, bedroom and bathroom counts, geographic coordinates, and categorical amenity flags (parking, laundry, pet policies, wheelchair access, smoking policy, electric vehicle charging, furnished status).

Constraints

- Dataset reflects 2020 market conditions; no real-time pricing data is available
- No GPU or TPU hardware available, which limits NLP analysis of the listing description
- Geographic scope is restricted to Southern states

Risks and Contingencies

- *Missing values:* public listing data commonly contains incomplete amenity fields; plan to exclude records with unusable location data.
- *Anomalous records:* listing platforms may include erroneous entries (e.g., zero-price listings, implausible square footage); plan to apply threshold-based filters before modeling
- *Outlier contamination:* regional price distributions in diverse Southern markets may be skewed by extreme listings; plan to apply outlier removal at the state/region level

Task 1.3: Determine Data-Mining Goals

The data-mining goals for this project are twofold:

- **Predictive goal:** predict apartment rental price using all available features in the dataset
- **Analytical goal:** determine the relative importance of each amenity feature in driving the predicted price, producing an actionable ranking for investors and property managers

Modeling Strategy

Three tree-based ensemble methods are used: Random Forest, Gradient Boosting, and XGBoost. Tree-based methods are selected because they handle mixed data types (numeric and categorical), are robust to outliers, and natively produce feature importance scores. A K-Means clustering step on latitude and longitude engineers a `location_cluster` feature to capture local market effects beyond state-level geography.

Success Metrics

Model performance is evaluated on a held-out 20% test set using: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R-squared (R^2), Average Squared Error (ASE), and Sum of Squared Errors (SSE).

Task 1.4: Produce a Project Plan

This project follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology across six structured phases: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment. Each phase produces one or more formal deliverables submitted according to the course schedule.

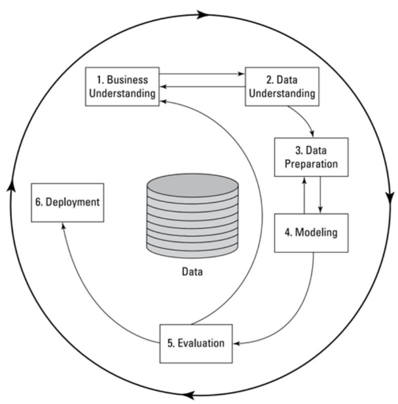

Figure 1: CRISP-DM pipeline diagram

Tools and Hardware

Category	Details
Language / IDE	Python 3.12.3, VS Code, Jupyter Notebook
Key Libraries	pandas 3.0.1, numpy 2.4.2, matplotlib 3.10.8, scipy 1.17.1
	scikit-learn 1.8.0, xgboost 3.2.0, category-encoders 2.9.0, joblib 1.5.3
Data Source	Kaggle (Craigslist scrape, January 2020)
Hardware	2 vCPU, 2 GB RAM, Ubuntu 24 Server

Complete Project Timeline

Task	Description	Submission
1.1	Determine Business Objectives	9 May 2026
1.2	Assess the Situation	9 May 2026
1.3	Determine Data-Mining Goals	9 May 2026
1.4	Produce a Project Plan	9 May 2026
2.1	Gather Data	23 May 2026
2.2	Describe Data	23 May 2026
2.3	Explore Data	23 May 2026
2.4	Verify Data Quality	23 May 2026
3.1	Select Data	6 Jun 2026
3.2	Clean Data	6 Jun 2026
3.3	Construct Data	6 Jun 2026
3.4	Integrate Data	6 Jun 2026
3.5	Format Data	6 Jun 2026
	Midterm Presentation	13 Jun 2026
4.1	Select Modeling Technique	20 Jun 2026
4.2	Generate Test Design	20 Jun 2026
4.3	Build Model	20 Jun 2026
4.4	Assess Model	20 Jun 2026
5.1	Evaluate Results	4 Jul 2026
5.2	Review Process	4 Jul 2026
5.3	Determine Next Steps	4 Jul 2026
6.1	Plan Deployment	11 Jul 2026
6.2	Plan Monitoring	11 Jul 2026
6.3	Final Report and Presentation	1 Aug 2026
6.4	Review Project	1 Aug 2026